Vehicle Orientation Estimation Using Conventional Image Processing

Group 6

Dhruvika Mehta
AU2440098
dhruvika.m@ahduni.edu.in

Krisha Desai
AU2440103
krisha.d@ahduni.edu.in

Manthan Pancholi
AU2440276
manthan.p1@ahduni.edu.in

Rim Shah
AU2440262
rim.s@ahduni.edu.in

***Abstract*—This project aims to estimate the orientation of a vehicle in aerial images using traditional image processing methods. The DRASHTI-HaOBB dataset provides aligned cubic box (OBB) labels, vehicle classes, class difficulty and heading angles. Only to localize vehicle crops, OBBs are used. The crop is used to estimate the orientation by means of Principal component analysis (PCA), probabilistic Hough Transform and a contour based minimum-area rectangle. A joint pipeline is employed for preprocessing, including grayscale conversion, contrast-limited adaptive histogram equalization (CLAHE), Gaussian smoothing, Canny edge detection and morphological closing. The mean angular errors were 10.74° for PCA, 12.46° for Hough and 10.97° for contour estimation over 100 different training images, in which a single difficulty-0 vehicle was selected for each image. PCA and contour returned valid estimates for all samples, Hough returned 98. The results provide a basic level for future research and the differences between the methods are small in comparison to the spread of errors.**

***Keywords*—vehicle orientation estimation, digital image processing, PCA, Hough Transform, contours, DRASHTI-HaOBB**

I. Introduction

Vehicle orientation is a useful geometric cue for aerial traffic analysis. DRASHTI-HaOBB consists of 27,577 annotated images and over 1.3 million vehicle instances from 14 classes with OBB coordinates and heading angles [1], [2]. Another computer vision problem that has been studied is vehicle orientation, including the learning-based approach for single images [3]. In this project, we examine classic approaches that calculate the orientation of a vehicle by analyzing edges and shape without training a deep model. The angular error is used to compare the three methods PCA, Hough Transform and contour geometry for the same crops. These methods are interpretable, do not require any training data and serve as a baseline to which learning-based estimators can be compared. This work brings two major contributions: a common preprocessing pipeline, and a like-for-like comparison of the three classical estimators on DRASHTI-HaOBB crops, as well as an error analysis by vehicle class.

II. Methodology

Dataset and ground truth: DRASHTI-HaOBB dataset consists of four oriented bounding box (OBB) corner points, vehicle class, difficulty level and heading angle per each annotation. The annotations were processed and matched with the images by their names. One of the 100 different training images (11 out of 14 classes) was chosen to evaluate one difficulty-0 vehicle. The distribution of classes in the samples selected was not balanced. Only the OBB coordinates were used to help identify the vehicle and to obtain an axis-aligned crop with 10% padding. Not accustomed to directly estimate the vehicle orientation.

The heading θ_h was converted to an undirected orientation in $[0^\circ, 90^\circ]$. With $r = \theta_h \bmod 180^\circ$,

$$\theta_{gt} = \min(r, 180^\circ - r). \tag{1}$$

This assumes that headings 180° apart are on the same axis. Each crop was subjected to 3 steps after gray level conversion: CLAHE (clip limit 2.0; 8×8 tiles), 5×5 Gaussian smoothing, median-based Canny edge detection and 3×3 morphological closing. The selection of the contours was made based on the area and the proximity to the crop centre.

PCA: Principal component analysis finds the direction of maximum variation among contour points. The first eigenvector (v_x, v_y) defines the axis, with angle

$$\theta_{\text{PCA}} = \operatorname{atan2}(v_y, v_x), \text{ normalized to } [0^\circ, 90^\circ] \tag{2}$$

Hough Transform: The probabilistic Hough Transform detects line segments in the image. Their angles θ_i are combined using segment lengths ℓ_i as weights. A doubled-angle mean accounts for the equivalence of θ and $\theta + 180^\circ$ [4]:

$$\hat{\theta}_H = \frac{1}{2} \operatorname{atan2}\left(\sum_i \ell_i \sin 2\theta_i, \sum_i \ell_i \cos 2\theta_i \right). \tag{3}$$

Contour method: OpenCV's minimum-area rotated rectangle (`minAreaRect`) is fitted to the selected contour, and the longaxis angle is used to estimate the angle. For prediction θ_p, angular error is $e = |\theta_p - \theta_{gt}|$.

III. Results

All three methods were tested on the same 100 vehicle samples. PCA and contour returned 100 valid estimates; Hough returned 98. Table I summarizes the overall errors. PCA had the lowest mean error (10.74°), followed by contour (10.97°)

and Hough (12.46°). Errors were within 10° for 74/100 PCA, 76/100 contour, and 71/98 valid Hough predictions.

TABLE I
OVERALL ANGULAR-ERROR RESULTS (DEGREES).

Method	Valid	Mean	Median	Std.	Max
PCA	100/100	10.74	4.85	16.11	81.71
Hough	98/100	12.46	4.18	20.44	82.39
Contour	100/100	10.97	5.72	16.44	89.00

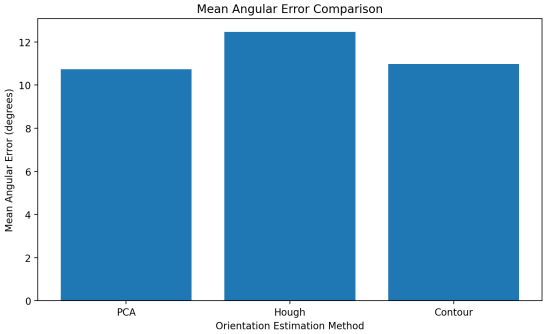


Fig. 1. Mean angular error for the three methods.

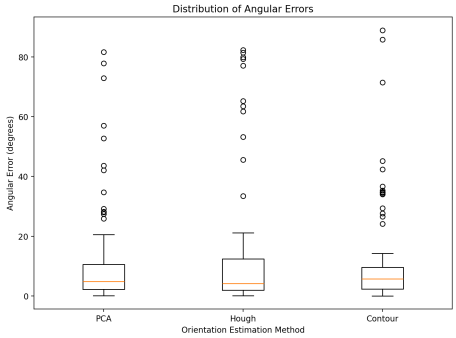


Fig. 2. Distribution of angular errors, including outliers.

IV. DISCUSSIONS

No method led on every metric. Their mean error was the lowest for PCA, and the lowest median for Hough, while the largest percentage of estimates were within 10° for contour. Hough also had the highest standard deviation (20.44°) and two failed predictions. Even though some crops are yet difficult for all three methods, the large maximum errors (81.71° for PCA, 82.39° for Hough, and 89.00° for contour) found.

Table II presents the results for each of the vehicle classes. The mean errors were lower for SUV and AutoRicksaw and higher for MotorCycle and Sedan for PCA, and the opposite was true for Hough. For Container vehicles the mean error is the lowest using the contour method, however only 3 samples are available for this class. These results are preliminary as the number of samples varied with each class.

TABLE II
CLASS-WISE MEAN ANGULAR ERROR (DEGREES).

Class	n	PCA	Hough	Contour
Auto3WCargo	3	11.34	38.58	13.41
AutoRicksaw	8	7.49	13.97	8.29
Bus	2	27.32	0.74	24.18
Container	3	40.12	49.13	27.49
MotorCycle	30	7.28	4.54	10.15
PickUp	4	8.73	20.72	3.86
SUV	34	7.64	12.97	8.34
Sedan	7	23.55	7.03	18.10
Tanker	1	0.19	3.27	7.00
Trailer	3	28.97	32.87	30.21
Truck	5	8.05	17.50	6.39

Possible causes of the large errors, such as background edges, fragmented contours and ambiguous vehicle shape, were not isolated in this study. The current evaluation uses only difficulty-0 vehicles from 100 training images and is class-imbalanced. Crop dimensions and Laplacian-variance sharpness were calculated during experimentation, but their relationship with error has not yet been analysed. Larger evaluation and analysis by vehicle size, image quality and orientation remain necessary.

V. CONCLUSION

A comparison of PCA, Hough Transform and contour based estimation was carried out on a common sample of 100 vehicles. PCA gave the lowest mean angular error, the lowest median error was given by Hough and the highest percentage of error of less than 10○ was given by Contour. No one method scored highest in each category. Before choosing a final method, there is a need for a more balanced and an evaluation and analysis of the vehicles by size, image quality and image orientation.

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